

Exercice 1 (3pts)

- 1) a) $n_0(S_2O_8^{2-}) = C_1 V_1 = 10 \text{ mmol}$
 $n_0(I^-) = C_2 V_2 = 15 \text{ mmol}$ 0,25
 b) $\frac{n_0(S_2O_8^{2-})}{2} > \frac{n_0(I^-)}{2} = 7,5 \cdot 10^3$ donc I^- est le réactif limitant.

$$n_{\infty}(I_2) = \frac{n_0(I^-)}{2} = 7,5 \text{ mmol}$$

$$[I_2]_{\infty} = \frac{7,5}{1} = 7,5 \text{ mmol L}^{-1} \quad \text{0,25}$$

- c) A l'état final $t \rightarrow +\infty$,

$$[I_2]_{\infty} = \lim_{t \rightarrow +\infty} [I_2] = a \Rightarrow a = 7,5 \text{ mmol L}^{-1} \quad \text{0,25}$$

2) a) $v = + \frac{d[I_2]}{dt}$ 0,15

$$b) v = \frac{d[I_2]}{dt} = \frac{a^2 k}{(1 + akt)^2} \quad \text{0,15}$$

c) A $t=0 \Rightarrow v_0 = a^2 k$ 0,15

$$d) [I_2] = a - \frac{a}{(1 + akt)}$$

En prenant $[I_2] = 4 \cdot 10^{-3} \text{ mmol/L}$; $a = 7,5 \text{ mmol L}^{-1}$
 $t = 20 \text{ min}$

$$4 \cdot 10^{-3} = 7,5 \cdot 10^{-3} - \frac{7,5 \cdot 10^{-3}}{(1 + 7,5 \cdot 10^{-3} \times 20 \times 60 \times k)}$$

les calculs donnent: $k = 0,127 \text{ L mol}^{-1} \text{ s}^{-1}$ 0,25

- A $t=0 \Rightarrow v_0 = 7,14 \cdot 10^{-6} \text{ mol L}^{-1} \text{ s}^{-1}$

soit $v_0 = 0,43 \text{ mmol L}^{-1} \text{ min}^{-1}$ 0,25

- A $t=20 \text{ min} \Rightarrow v_{20} = 1,55 \cdot 10^{-6} \text{ mol L}^{-1} \text{ s}^{-1}$

soit $v_{20} = 0,09 \text{ mmol L}^{-1} \text{ min}^{-1}$

- 3) a) Détermination graphique.

$$v_{t=0} = \frac{5-0}{12-0} = 0,42 \text{ mmol L}^{-1} \text{ min}^{-1} \quad \text{0,25}$$

$$v_{t=20} = \frac{5,8-4}{40-20} = 0,09 \text{ mmol L}^{-1} \text{ min}^{-1}$$

b) $\frac{v(I^-)}{2} = \frac{v(I_2)}{1} \Rightarrow v(I^-) = 2v(I_2)$

$$v_{20}(I^-) = 2 \times 0,09 = 0,18 \text{ mmol L}^{-1} \text{ min}^{-1} \quad \text{0,25}$$

- c) temp de $1/2$ réaction: durée au bout de laquelle la moitié de l'avancement volumique maximal est atteint.

$$A t = t_{1/2} \Rightarrow [I_2] = \frac{[I^-]_0}{2} = 7,5 \text{ mmol}$$

D'après l'équation: $\frac{[I^-]_0}{2} = [I_2]_{1/2}$

$$\Rightarrow [I_2]_{1/2} = \frac{7,5}{2} = 3,75 \text{ mmol L}^{-1} \text{ min}^{-1}$$

Graphiquement on a: $t_{1/2} = 17,5 \text{ min}$ 0,15



$t=0$	10 mmol/L	15 mmol/L	0	0
t	$(10-x_v)$	$(15-2x_v)$	$2x_v$	x_v

A $t=20 \text{ min} \Rightarrow [I_2] = 4 \text{ mmol L}^{-1} = x_v$

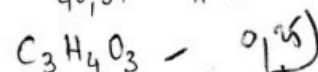
$$\Rightarrow [I^-] = 15 - 2x_v = 7 \text{ mmol L}^{-1}$$

$$\Rightarrow [SO_4^{2-}] = 2x_v = 8 \text{ mmol L}^{-1} \quad \text{0,15}$$

$$\Rightarrow [S_2O_8^{2-}] = 10 - x_v = 6 \text{ mmol L}^{-1}$$

Exercice 2 (3pts)

1) $C_x H_y O_3$
 $\frac{88}{100} = \frac{12x}{40,91} = \frac{y}{4,54} = \frac{16z}{54,55}$



a) $A_2 \Rightarrow$ acide carboxylique

$A_1 \Rightarrow$ acide carboxylique

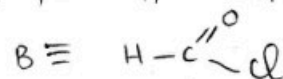
$A \Rightarrow$ ~~diacide~~ (anhydride diacide)

$A \Rightarrow$ anhydride d'acide 0,15

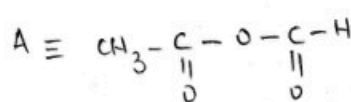
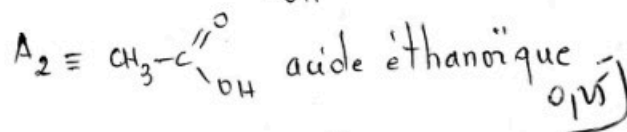
$B \Rightarrow$ chlorure d'acyle

b) $B = C_n H_{2n+1} - C(=O)Cl$

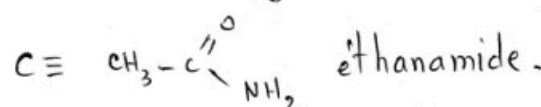
$$M_B = 14n + 64,5 = 64,5 \Rightarrow n = 0$$

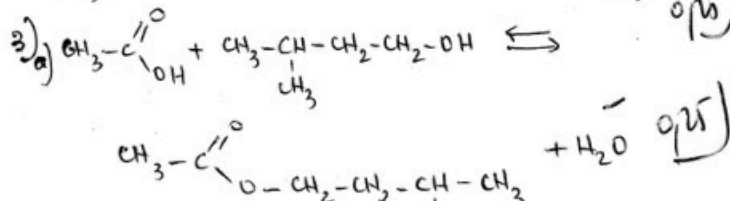
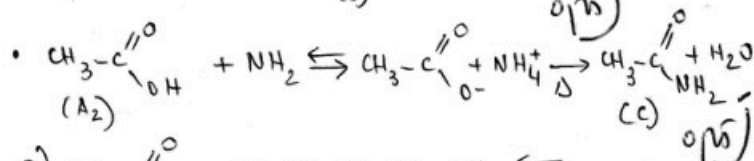
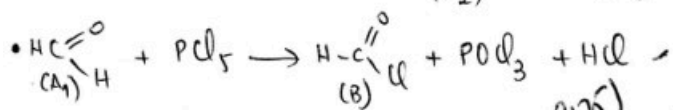
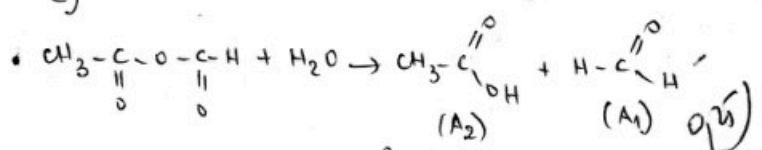


donc $A_1 \equiv H - C(=O)OH$ acide méthanoïque 0,15



anhydride éthanoïque et méthanoïque 0,15

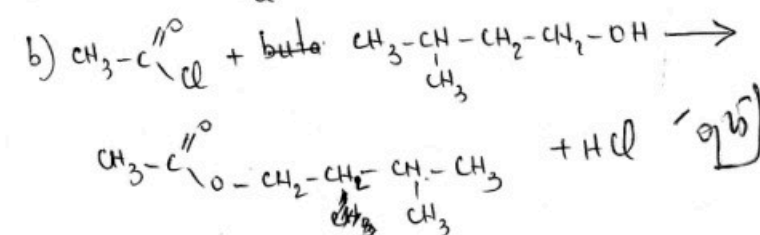




b) (D): éthanoate de 3-méthylbutyle (ester) - 0,75

c) elle est lente et limitée. - 0,75

4) a) $\text{CH}_3-\text{C}(=\text{O})-\text{Cl}$ chlorure d'éthanoyle - 0,75



Exercice 3

I. 1) MC Uniforme de rayon $r = R \cos \lambda$ - 0,75

$$2) v_0 = r \omega_0 = R \cos \lambda \omega_0 = \frac{2\pi R \cos \lambda}{T_0}$$

$$E_{c0} = \frac{1}{2} m v_0^2 = \frac{2m\pi^2 R^2 \cos^2 \lambda}{T_0^2} \quad \text{0,75}$$

$$3) E_0 = E_{p0} + E_{c0} = -\frac{GMm}{R} + \frac{2m\pi^2 R^2 \cos^2 \lambda}{T_0^2}$$

$$E_{c0} = 5,11 \cdot 10^7 \text{ J}$$

$$\text{AN: } v_0 = 3,57 \cdot 10^2 \text{ m s}^{-1}$$

$$E_0 = -5 \cdot 10^{10} \text{ J} \quad \text{0,75}$$

$$\text{II - 1) a) } \vec{F} = -\frac{GMm}{r^2} \vec{u} = m \vec{a}$$

$$\vec{a} = -\frac{GMm}{r^2} \vec{u} = \frac{GMm}{r^2} \vec{n} = \frac{v^2}{r} \vec{n}$$

$$v = \sqrt{\frac{GM}{r}} \quad \text{avec } g_0 = \frac{GM}{R^2}$$

$$\Rightarrow GM = g_0 R^2 \text{ d'où } v = \sqrt{\frac{g_0 R^2}{r}} \quad \text{0,75}$$

$$b) T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r^3}{g_0 R^2}} \quad \text{0,75}$$

$$c) E_c = \frac{1}{2} m v^2 = \frac{m g_0 R^2}{2r}; E_p = -\frac{m g_0 R^2}{r}$$

$$E = E_p + E_m = -\frac{g_0 m R^2}{2r} \quad \text{0,75} \quad (2)$$

$$2) a) v_1 = \sqrt{\frac{g_0 R^2}{R}} = \sqrt{g_0 R} \quad \text{0,75}$$

$$b) T_1 = 2\pi \sqrt{\frac{R^3}{g_0 R^2}} = 2\pi \sqrt{\frac{R}{g_0}} \quad \text{0,75}$$

$$E_1 = -\frac{m g_0 R^2}{2R} = -\frac{m g_0 R}{2} \quad \text{0,75}$$

$$c) g_0 = \frac{GM}{R^2} = 9,77 \text{ m s}^{-2}$$

$$v_1 = \sqrt{g_0 R} = 7,9 \cdot 10^3 \text{ m s}^{-1} \quad \text{0,75}$$

$$E_1 = -\frac{m g_0 R^2}{2} = -2,5 \cdot 10^{10} \text{ J}$$

$$T_1 = 5085,37 \text{ s}$$

$$d) \mathcal{E} = \Delta E = E_1 - E_0$$

$$\mathcal{E} = -\frac{m g_0 R^2}{2} - \left(-\frac{GMm}{R} + \frac{2\pi^2 m R^2 \cos^2 \lambda}{T_0^2} \right)$$

$$\mathcal{E} = \frac{m g_0 R}{2} - \frac{2m\pi^2 R^2 \cos^2 \lambda}{T_0^2} \quad \text{0,75}$$

Cette énergie dépend du point de lancement. Les bases de lancement les plus favorables du point de vue énergétique se situent le plus près possible de l'équateur ($\lambda = 0$). 0,75

3) a) immobile par rapport à un point de la Terre. Il a la même période que la Terre: $T_2 = 1 \text{ j sidéral} = 86164 \text{ s}$. 0,75

$$b) T_2 = 2\pi \sqrt{\frac{r_2^3}{g_0 R^2}}$$

$$\Rightarrow r_2 = \left(\frac{T_2^2 g_0 R^2}{4\pi^2} \right)^{1/3} = 4,21 \cdot 10^7 \text{ m} \approx 42100 \text{ km} \quad \text{0,75}$$

$$\Rightarrow z_2 = r_2 - R = 35700 \text{ km}$$

$$c) v_2 = \sqrt{\frac{g_0 R^2}{r_2}} = 3,08 \text{ km s}^{-1} \quad \text{0,75}$$

$$E_2 = -\frac{m g_0 R^2}{2r_2} = -3,8 \cdot 10^9 \text{ J}$$

$$d) T' = 2\pi \sqrt{\frac{r_2^3}{g_0 R^2}} = 28093 \text{ s}$$

$$\omega_2' = \frac{2\pi}{T'} = 2,236 \cdot 10^{-4} \text{ rad s}^{-1} \quad \text{0,75}$$

$$\frac{r^2}{r^3} = \frac{T'^2}{r'^3} \text{ loi de Kepler.}$$

$$(S_1) \Rightarrow \omega_2 = \frac{\theta_2}{\tau} \Rightarrow \theta_2 = \omega_2 \tau$$

$$(S_2) \Rightarrow \omega' = \frac{\theta'}{\tau} \Rightarrow \theta' = \omega' \tau$$

$$\theta'_2 - \theta_2 = 2\pi = \tau(\omega' - \omega_2)$$

$$\tau = \frac{2\pi}{\omega' - \omega_2} = \frac{2\pi}{2,236 \cdot 10^4 - 7,29 \cdot 10^5}$$

$$\tau = 41683,5 \text{ s} - 0,5$$

Exercice 4 (4pts)

$$1) a) \frac{1}{2} m v_B^2 = mgAB \sin \alpha$$

$$v_B = \sqrt{2gAB \sin \alpha} = 4 \text{ m/s} - 0,5$$

$$\frac{1}{2} m v_c^2 - \frac{1}{2} m v_B^2 = mgr(1 - \cos \alpha)$$

$$v_c = \sqrt{v_B^2 + 2gr(1 - \cos \alpha)} = 4,29 \text{ m/s} - 0,5$$

$$\frac{1}{2} m v_D^2 - \frac{1}{2} m v_c^2 = -mgr(1 - \cos \delta) \text{ avec } \delta = 60^\circ$$

$$v_D = \sqrt{v_c^2 - 2gr(1 - \cos \delta)} = 3,07 \text{ m/s} - 0,5$$

$$b) \text{ Enc: } \vec{P} + \vec{R}_c = m \vec{a}_c$$

$$-P + R_c = m \frac{v_c^2}{r} \Rightarrow R_c = m \left(g + \frac{v_c^2}{r} \right) = 1,52 \text{ N} - 0,5$$

$$\text{En D: } \vec{P} + \vec{R}_D = m \vec{a}_D$$

$$-P \sin \alpha + R_D = m \frac{v_D^2}{r} \Rightarrow R_D = m \left(g \sin \alpha + \frac{v_D^2}{r} \right)$$

$$R_D = 0,77 \text{ N} - 0,5$$

$$e) \vec{v}_D \begin{cases} \cdot \text{pt: D} \\ \cdot \text{direction: faisant } 60^\circ \text{ avec } OX \\ \cdot \text{sens: vers le haut} \\ \cdot \text{norme: } v_D = 3,07 \text{ m/s} \end{cases} - 0,5$$

$$2) \vec{a} \begin{cases} \dot{x} = 0 \\ \dot{y} = -g \end{cases} \Rightarrow \vec{v} \begin{cases} x = v_D \cos \delta \\ y = -gt + v_D \sin \delta \end{cases}$$

$$O\vec{n} = \begin{cases} x = v_D t \cos \delta \\ y = -\frac{1}{2} g t^2 + v_D t \sin \delta + h + r(1 - \cos \delta) \end{cases}$$

$$y = -\frac{g}{2v_D^2 \cos^2 \delta} x^2 + x \tan \delta + h + r(1 - \cos \delta)$$

$$y = -2,122 x^2 + 1,732 x + 2 - 0,5$$

$$b) \text{ Au sommet } y' = 0 \Rightarrow x_F = 0,408 \text{ m}$$

$$H = y(x_F) = 2,35 \text{ m} - 0,5$$

$$c) \text{ résoudre } y(x) = 0$$

$$x = 1,46 \text{ m} \text{ et } x = -0,64 \text{ m} < 0 \text{ (à rejeter)}$$

$$\text{on trouve: } OP = 1,46 \text{ m} - 0,5$$

$$d) x_P = v_D t_P \cos \delta \Rightarrow t_P = \frac{x_P}{v_D \cos \delta} = 0,951 \text{ s}$$

$$\vec{v}_P \begin{cases} v_{Px} = v_D \cos \delta = 1,535 \text{ m/s} \\ v_{Py} = -gt_P + v_D \sin \delta = -6,85 \text{ m/s} \end{cases}$$

$$v_P = \sqrt{v_{Px}^2 + v_{Py}^2} = 7,02 \text{ m/s} - 0,5$$

$$\tan \delta = \frac{v_{Py}}{v_{Px}} \Rightarrow \delta = -77,4^\circ - 0,5$$

$$e) y = \tan \theta x = 0,278 x$$

$$y = -2,122 x^2 + 1,732 x + 2$$

Point d'intersection d'abscisse positive -

$$P'(1,37 \text{ m}; 0,38 \text{ m}) - 0,5$$

Exercice 5

$$1) J_D = J_G + 2(J_S + md^2)$$

$$J_D = \frac{1}{2} MR^2 + 2m \left(\frac{2}{5} r^2 + d^2 \right) = 4,94 \cdot 10^{-2} \text{ kg m}^2 - 1$$

$$2) \sum M = J_D \ddot{\theta} \Rightarrow M_m = J_D \ddot{\theta} \Rightarrow \ddot{\theta} = \frac{M_m}{J_D} = \text{cte}$$

$$\ddot{\theta} = \frac{\Delta \omega}{\Delta t} = \frac{\omega}{\Delta t} = \frac{M_m}{J_D} \Rightarrow M_m = J_D \frac{\omega}{\Delta t}$$

$$M_m = \frac{J_D \omega}{\Delta t} = 9,88 \cdot 10^2 \text{ Nm} - 1$$

$$3) \text{ système: une sphère.}$$

$$\text{TCF: } \vec{R} + \vec{P} + \vec{T} = m \vec{a}$$

$$(n) T = m a_n = \frac{m v^2}{\ell} = m \ell \omega^2 = k \Delta \ell$$

$$\text{avec } \ell = 8 + 2 = 10 \text{ cm}$$

$$k = \frac{m \ell \omega^2}{\Delta \ell} = 25 \text{ N/m} - 1$$